

4.4. $U_4 = \neg p \vee (\neg q \vee r) \rightarrow q \vee \neg p \vee r$
Find the anti-models (sem. tab.)

Seminar 9
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The anti-models of U are the models of $\neg U$.

$$\neg U_4 = \neg (\neg p \vee (\neg q \vee r) \rightarrow q \vee \neg p \vee r) \quad (1)$$

| α for (1).

α rules

$A \wedge B$

|

A

|

B

$\neg(A \rightarrow B)$

|

A

|

$\neg B$

β rules

$A \vee B$

A B

$A \rightarrow B$

$\neg A$ B

$$\neg U_4 = \neg (\neg p \vee (\neg q \vee r) \rightarrow q \vee \neg p \vee r) \quad (1)$$

| α for (1)

$$\neg p \vee (\neg q \vee r) \quad (2)$$

|

$$\neg(q \vee \neg p \vee r) = \neg q \wedge p \wedge \neg r \quad (3)$$

| α for (3)

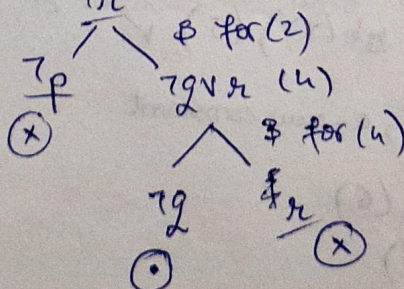
$\neg q$

|

p

|

$\neg r$



$$\text{DNF}(\neg U_4) = (\underbrace{\neg q \wedge p \wedge \neg r \wedge \neg p}_{\equiv \text{F}}) \vee (\underbrace{\neg q \wedge p \wedge \neg r}_{\equiv \text{F}}) \vee (\underbrace{\neg q \wedge p \wedge \neg r \wedge r}_{\equiv \text{F}}) \quad (1)$$

Cube $\neg g \wedge p \wedge r \equiv T$ provides 1 model of $\neg U_4$:

$$i: \{p, q, r\} \rightarrow \{T, F\}$$

$$i(p) = T$$

$$i(q) = F$$

$$i(r) = F$$

$$i(\neg U_4) = T$$

$$i(U_4) = F.$$

i antimodel of U_4 .

$$(5) \quad H_1, H_2, H_3 \models c$$

$$H_1: (\forall x) (hb(x) \rightarrow rc(x))$$

$$H_2: (\forall x) (\neg hb(x) \rightarrow \neg lh(x))$$

$$H_3: (\forall x) (\neg lh(x) \rightarrow \neg rc(x))$$

$$c: (\forall x) (hb(x) \rightarrow \neg hb(x))$$

(T) $H_1, H_2, H_3 \models c$ iff $H_1 \wedge H_2 \wedge H_3 \wedge \neg c$ has a closed table.

$$H_1 \wedge H_2 \wedge H_3 \wedge \neg c \quad (1)$$

| α rule (1)

$$H_1: (\forall x) (hb(x) \rightarrow rc(x)) \quad (2) \checkmark$$

|

$$H_2: (\forall x) (\neg hb(x) \rightarrow \neg lh(x)) \quad (3)$$

|

$$H_3: (\forall x) (\neg lh(x) \rightarrow \neg rc(x)) \quad (4)$$

|

$$\neg c: (\exists x) (hb(x) \wedge \neg hb(x)) \quad (5) \checkmark$$

| δ rule (5), a-new constant

$$hb(a) \wedge \neg hb(a) \quad (6)$$

| α rule (6)

$$hb(a)$$

$$\neg hb(a)$$

$\exists$ rule:

$$(\exists x) A(x)$$

|

$A(c)$, c - new const.

$\forall$ rule:

$$(\forall x) A(x)$$

|

$$A(c_1)$$

|

...

|

$A(c_m)$

|

$$(\forall x) A(x) \text{ copy}$$

c_1

...

c_m

↓

all the constants on the branch

(2)

| $\exists$ rule (2) $\rightarrow$ a used for instantiation.

$hb(a) \rightarrow hc(a)$ (7)

(2) copy

$\exists$ rule (7)

$\neg hb(a)$

(x)

$hc(a)$

| $\exists$ rule (3), a used for instantiation.

$\neg sb(a) \rightarrow \neg lh(a)$ (8)

(3) copy

$\exists$ rule (8)

$hb(a)$

$\neg lh(a)$ (9)

(x)

| $\exists$ rule (4), a used for instantiation

$\neg lh(a) \rightarrow \neg hc(a)$ (9)

(4) copy

$\exists$ rule (9)

$lh(a)$

(x)

$\neg hc(a)$

(x)

Conclusion: $H_1 \wedge H_2 \wedge H_3 \wedge \neg C$ has a closed semantic tableau with 4 closed branches $hb(a), \neg hb(a)$

$sb(a), \neg sb(a)$

$lh(a), \neg lh(a)$

$hc(a), \neg hc(a)$

So $H_1 \wedge H_2 \wedge H_3 \wedge \neg C$ is inconsistent and $H_1, H_2, H_3 \models C$.

ex. 6.4 '∀' is semi-distributive over '→'.

$$a) \models \underbrace{(\forall x)(A(x) \rightarrow B(x)) \rightarrow ((\forall x)A(x) \rightarrow (\forall x)B(x))}_{U_1}$$

$\models U_1$ iff $\neg U_1$ has a closed semantic table.

$$\neg U_1 = \neg ((\forall x)(A(x) \rightarrow B(x)) \rightarrow (\forall x)A(x) \rightarrow (\forall x)B(x)) \quad (1) \checkmark$$

=
| $\neg$ rule for (1)

$$(\forall x)(A(x) \rightarrow B(x)) \quad (2) \checkmark$$

$$\neg ((\forall x)A(x) \rightarrow (\forall x)B(x)) \quad (3) \checkmark$$

| $\neg$ rule for (3).

$$(\forall x)A(x) \quad (4) \checkmark$$

$$\neg (A(x)B(x)) \equiv \neg (\forall x) \neg (\exists x) \neg B(x) \quad (5) \checkmark$$

| $\exists$ rule for (5), c = new constant.

$$\neg B(c)$$

| $\exists$ rule for (4), c used for instantiation.

$$A(c)$$

|
(4) copy

| $\exists$ rule for (2), c used for instantiation.

$$A(c) \rightarrow B(c) \quad (6)$$

|
(2) copy

$\vee$ rules for (6).

$$\neg A(c)$$

(X)

$$B(c)$$

(X)

Conclusion: $\neg U_1$ has a closed semantic table with 2 closed branches $(\neg A(c), A(c))$ and $(\neg B(c), B(c))$, so $\neg U_1$ is inconsistent and $\models U_1$ (U_1 is a tautology)

$$6) \neg U_2 = \neg ((\forall x) A(x) \rightarrow (\forall x) B(x)) \rightarrow (\forall x) (A(x) \rightarrow B(x)) \quad (1) \checkmark$$

| α rules for (1)

$$(\forall x) A(x) \rightarrow (\forall x) B(x) \quad (2)$$

|

$$\neg (\forall x) (A(x) \rightarrow B(x)) \quad (3) \checkmark$$

$\equiv$

$$(\exists x) (\neg (A(x) \rightarrow B(x)))$$

| $\exists$ for (3), c new constant

$$\neg (A(c) \rightarrow B(c)) \quad (4)$$

| α rule for (4)

two different constants
 $\Rightarrow$ opened branch.

$$\underline{A(c)}$$

$$\neg B(c)$$

| β rule for (2)

$$\neg (\forall x) A(x) \quad (5)$$

$$(\forall x) B(x) \quad (6)$$

$\equiv$

$$(\exists x) \neg A(x)$$

| $\exists$ rule for (5)
 k new constant

| $\exists$ rule for (6)

c constant used for instantiation

$$B(c)$$

|
(6) copy

$$\underline{\neg A(k)}$$



Conclusion:

$\neg U_2$ has a finite, complete, and open semantic table, having one closed branch and one opened branch.

$$\hookrightarrow B(c), \neg B(c)$$

$\neg U_2$ is consistent and $U_2 \not\models$

$I = U_1$ and $I \neq U_2$, so $(\forall)$ is semi-distributive over implication.

! (open branch gives the F val. of... because $U_2 \not\models$.)

Ex1. Predicate res.

Prove that the following formulas are theorems, using resolution.

$$U_4 = (A \rightarrow c) \rightarrow ((\neg A \rightarrow B) \rightarrow (\neg B \rightarrow c)).$$

Th. result:

$$\vdash u \text{ iff } \text{CNF}(\neg u) \vdash_{\text{res}} \square.$$

$$\boxed{A \rightarrow B \equiv \neg A \vee B}$$

$$\neg \vee l, g \vee \neg l \vdash_{\text{res.}} \neg \vee g$$

$$\text{Res}(\neg \vee l, g \vee \neg l) = \neg \vee g$$

$$\neg U_4 = \neg ((A \xrightarrow{1} c) \xrightarrow{2} ((\neg A \xrightarrow{3} B) \xrightarrow{4} (\neg B \xrightarrow{5} c)))$$

Apply normalisation algorithm:

$$\neg U_4 \xrightarrow[\text{replace } \xrightarrow{1, 3, 5}]{=} \neg ((\neg A \vee c) \xrightarrow{2} (A \vee \neg B) \xrightarrow{4} (B \vee \neg c))$$

$$\xrightarrow[\text{replace } \xrightarrow{4}]{=} \neg ((\neg A \vee c) \rightarrow (\neg A \vee B) \vee (B \vee c))$$

$$\xrightarrow[\text{replace } \xrightarrow{3}]{=} \neg (\neg (\neg A \vee c) \vee (\neg A \vee B) \vee (B \vee c))$$

$$\xrightarrow[\text{DeMorgan}]{=} (\underbrace{\neg \neg A}_{c_1} \vee \underbrace{\neg c}_{c_2}) \wedge (\underbrace{\neg A}_{c_3} \vee \underbrace{B}_{c_4}) \wedge (\underbrace{B}_{c_3} \vee \underbrace{\neg c}_{c_4})$$

$$\Rightarrow \text{CNF}(\neg U_4).$$

$$\{c_1, c_2, c_3, c_4\} \vdash_{\text{res}}^? \square$$

$$c_5 = \text{Res}_B(c_2, c_3) = A$$

$$c_6 = \text{Res}_A(c_1, c_5) = C$$

$$c_7 = \text{Res}_C(c_6, c_4) = \square$$

Conclusion: From $\text{CNF}(\neg U_4)$ we derive $\square$

$$\text{CNF}(\neg U_4) \vdash_{\text{res}} \square \Rightarrow \neg U_4 \text{ inconsistent} \\ \Rightarrow U_4 \text{ valid (theorem)} \\ \leftarrow U_4 \leftarrow$$

(6)